

POTENTIOMETER (2)

Aims

- A. To compare electromotive forces (emfs) of Leclanché and Daniel cells
- B. To determine the internal resistance of Leclanché cell.

Materials and Apparatus

Potentiometer, Daniel cell, Leclanché cell, power supply, two resistance boxes, galvanometer, sliding key, two-way switch, two keys, and connecting wires.

Theory

Part A:

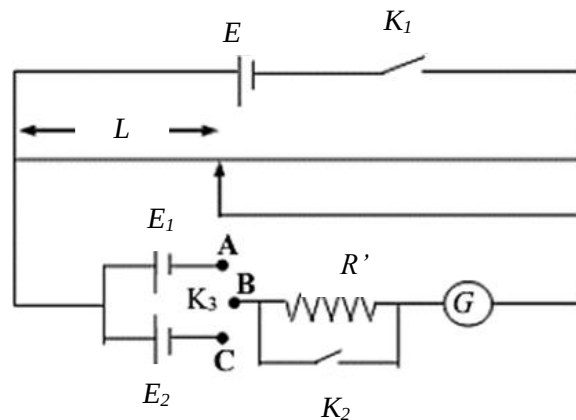


Fig. 1: The experimental circuit to compare the emf's of two cells
 Where, E_1 : Leclanché cell of emf E_1 , E_2 : Daniel cell of emf E_2 , E : power supply of voltage E , G : galvanometer, R' : resistor of higher resistance, K_1, K_2 : keys, K_3 : two way switch, L : balance length

At the balance point of the setup shown in Fig.1,

when A and B are connected

$$E_1 = k L_1 \quad (1)$$

when C and B are connected

$$E_2 = k L_2 \quad (2)$$

Where,

k : potential drop per unit length of the potentiometer wire

From (1) and (2),

$$\frac{E_1}{E_2} = \frac{L_1}{L_2} \quad (3)$$

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Part B:

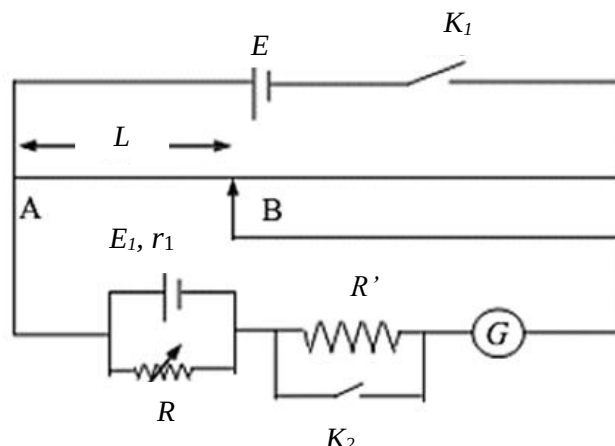


Fig. 2: The experimental circuit to find the internal resistance
Where, R :variable resistor (of resistance R)

For the circuit shown in Fig.2, at the balance point, $V_{AB} = IR = kL$ and $E_1 = I(R + r_1)$

Where,

r_1 : internal resistance of the cell

Therefore,

$$\frac{1}{L} = \frac{kr_1}{E_1} \frac{1}{R} + \frac{k}{E_1} \quad (4)$$

Procedure

Part A:

- a) Setup the circuit as shown in Fig. 1
- b) Using the two-way switch connect one of the given cells and obtain the balance length L_1 (connect AB or BC).
- c) Repeat the same procedure for the other cell to obtain L_2 .

Part B:

- a) Setup the circuit as shown in Fig. 2 using the Leclanché cell.
- b) Obtain a set of six values of balance length L for six suitable values of R .
- c) Plot the graph of $1/L$ vs. $1/R$ and find the internal resistance of the cell.